

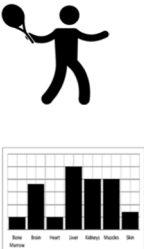
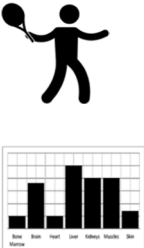
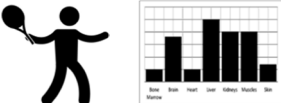
1.3.a. Biomechanical principles, levers and the use of technology

Learners will be able to develop their knowledge and understanding of the underlying biomechanical principles related to Newton's Laws and force, including the factors affecting air resistance and how this knowledge is applied to sports performance.

Learners will be able to calculate force, momentum, acceleration and weight.

The components of a lever system will be known for 1st, 2nd and 3rd class levers.

Learners will also develop their knowledge and understanding of the use of technology to analyse movement and improve performance.

Topic Area	Content
Biomechanical principles 	- Define and apply Newton's laws of motion: - Newton's first law: inertia - Newton's second law: acceleration - Newton's third law: reaction - Force: - net force - balanced and unbalanced force - weight - reaction - friction - air resistance - factors affecting friction and air resistance and their manipulation in sporting performance - free body diagrams showing vertical and horizontal forces acting on a body at an instant in time and the resulting motion - calculations of force, momentum, acceleration and weight - definition of centre of mass - factors affecting the position of the centre of mass - the relationship between centre of mass and stability.
Levers 	- components of a lever system: - load - effort - fulcrum - effort arm - load arm 1st class lever 2nd class lever 3rd class lever - mechanical advantage of a 2nd class lever.
Analysing movement through the use of technology 	- definitions and uses of: - limb kinematics - force plates - wind tunnels - how each type of technology may be used to optimise performance in sport.